

The Effect of Age on Functional Outcome in Mild Traumatic Brain Injury: 6-Month Report of a Prospective Multicenter Trial

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Objective: Elderly patients (aged 60 years and older) have been demonstrated to have an increased mortality after isolated traumatic brain injury (TBI); however, the prognosis of those patients surviving their hospitalization is unknown. We hypothesized that surviving elderly patients would also have decreased functional outcome, and this study examined the functional outcome of patients with isolated TBI at discharge and at 6 months posthospitalization.

Methods: This was a multicenter prospective study of all patients with isolated moderate to severe TBI defined as Head Abbreviated Injury Scale score of 3 with an Abbreviated Injury Scale score in any other body area of 1. Patients surviving to discharge gave their consent and were enrolled. Data collected included demographics, Glasgow Coma Scale (GCS) score at admission, and neurosurgical interventions. Outcome data included discharge disposition and Glasgow Outcome Scale score and modified Functional Independence Measure (FIM) score at discharge and at 6 months.

Results: Two hundred thirty-five patients were enrolled, with 44 (19%) aged

greater than or equal to 65 years. Mechanisms of injury were falls (34%), assaults (28%), motor vehicle collisions (14%), pedestrian (11%), and other (12%). Falls were more common in the older patients and assaults in the younger group. The mean admitting GCS score was 12.8 (95% confidence interval [CI], 12.4–13.3), with older patients having a higher mean GCS score, 14.1 (95% CI, 13.6–14.6) versus 12.5 (95% CI, 12.0–13.1; $p = 0.03$). There were no differences in the percentage of patients admitted to the intensive care unit or requiring neurosurgical intervention between younger and older patients. Because there were few elderly patients with low GCS scores who survived to discharge, outcome measures focused on those patients with GCS scores of 13 to 15. A greater percentage of elderly were discharged to rehabilitation (28% vs. 16%, $p = 0.08$). The mean discharge FIM score was 10.4 (95% CI, 9.8–11.0) for the elderly versus 11.4 (95% CI, 11.1–11.7) for the young ($p = 0.001$), with 68% elderly and 89% young discharged with total independent scores of 11 to 12. At 6 months, the difference narrowed, but the mean FIM

score was still greater for the young group, 11.7 (95% CI, 11.6–11.9) versus 11.0 (95% CI, 10.6–11.4; $p < 0.001$).

Conclusion: Functional outcome after isolated mild TBI as measured by the Glasgow Outcome Scale and modified FIM is generally good to excellent for both elderly and younger patients. Older patients required more inpatient rehabilitation and lagged behind their younger counterparts but continued to recover and improve after discharge. Although there were statistically significant differences in the FIM score at both discharge and 6 months, the clinical importance of these small differences in the mean FIM score to the patient's quality of life is less clear. Measurable improvement in functional status during the first 6 months after injury is observed in both groups. Aggressive management and care of older patients with TBI is warranted, and efforts should be made to decrease inpatient mortality. Continued follow-up is ongoing to determine whether these outcomes persist at 12 months.

Key Words: Traumatic brain injury, Elderly, Functional outcome, Mortality.

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Traumatic brain injury (TBI) is one of the most significant public health problem faced by trauma centers.^{1–4} The devastating impact of TBI on the patient, their family, and society from cost and resource use standpoints cannot be underestimated. In a recent analysis, the annual incidence of TBI in the United States was estimated to be 103 per 100,000

population. In male patients, the incidence peaks between the ages of 15 and 24 (248 per 100,000) and again above 75 years of age (243 per 100,000). The incidence in female patients peaks in the same age groups, but the absolute rates are lower (102 and 155, respectively). These rates underestimate the true incidence of head trauma, because patients with milder symptoms at the time of injury are usually not hospitalized.² This later peak in head injury is of increasing significance, as it estimated from the 2000 U.S. Census that there will be approximately 46 million people over 65 years of age by the year 2015 and over 11 million over the age of 80.

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Approximately three quarters of traumatic brain injuries that require hospitalization are nonfatal. However, each year, approximately 80,000 survivors of TBI will incur some disability or require prolonged medical care.² Age clearly appears to be an important determinant of survival after TBI.^{5,6} The mortality rate is higher for all levels of injury^{5,7} but increases particularly in the oldest old and in severe TBI, defined as those patients presenting with a Glasgow Coma Scale (GCS) score below 8.^{8–10} The effect of age on functional recovery is less certain.^{11,12}

Cifu and colleagues in a report from the TBI Model Systems matched 50 younger (age 18–50 years) with 50 older (age >55 years) patients admitted to acute care rehabilitation after TBI. The older patients were admitted with lower Functional Independence Measure (FIM) and disability rating scores. Although their stay in rehabilitation was longer and more costly, over 82% of older patients eventually were discharged to independent living. In a severely brain injured population, Kilaru et al. reported that older patients (>65 years) had high in-hospital mortality and poor functional recovery.¹¹ A serious limitation to their conclusion was that only 11 patients (27% of the study group) were alive at 3 years. Thus, it may be possible that should a patient survive their TBI, their recovery pattern is similar regardless of age.

Some investigators^{13,14} have proposed age as a significant predictor of functional outcome on the basis of studies that found the elderly are less likely than younger adults to live or function independently after TBI.^{7,11,13} Others have noted that the elderly take longer to resolve coma,¹⁵ require longer rehabilitation, and have a slower rate of change in functional measures, but will ultimately achieve a similar long-term functional outcome than their younger counterparts^{12,16,17} and can return to independent living.¹⁸ A serious deficiency is that the majority of studies have been retrospective, primarily based on inpatient rehabilitation populations and have included only small numbers of elderly survivors. Lastly, the impact of TBI on acute care rendered to older victims of multisystem injuries and the presence of multisystem injuries in patients with TBI remains to be fully elucidated, and these extracranial injuries are likely to confound any interpretation of the long-term outcome of TBI. Therefore, this study was designed to determine how age affects long-term functional outcome from isolated TBI, and we conducted a prospective, observational, multicenter study of adult patients who survived TBI at four Level I trauma centers. All patients who survived hospitalization for isolated TBI, regardless of rehabilitation therapy, were prospectively enrolled at discharge from the acute care hospital and reevaluated 6 months later.

PATIENTS AND METHODS

The study was conducted at four Level I trauma centers: New Jersey Trauma Center at University of Medicine and Dentistry of New Jersey–University Hospital, New Jersey Medical School, Newark, New Jersey, University of Califor-

nia at San Diego, San Diego, California, San Francisco General Medical Center of the University of California, San Francisco, California, and Harborview Medical Center, University of Washington, Seattle, Washington. Institutional review board approval was obtained for the study at each institution enrolling patients and all patients were enrolled only after appropriate consent was obtained.

Adult patients (age > 18 years) admitted to one of the four trauma centers between January 2000 and July 2002 were eligible for enrollment if they sustained blunt TBI with head Abbreviated Injury Scale (AIS) score greater than 3 and an AIS score to any other body area of less than 1 and they survived to discharge. Patients who had minor facial or orbital fractures (AIS score of 2) that did not require operative fixation were also included in the study, as the consensus of the investigators felt that these injuries would have minimal impact on functional outcome. A head AIS score of 3 in this study included all patients with a brain injury seen on computed tomographic scanning such as subdural hematomas or intracerebral contusions. Patients with prolonged loss of consciousness without a definite injury on computed tomographic scanning were not included. In-hospital data collected included demographics, GCS score, mechanism and type of injury, neurosurgical intervention (craniotomy, placement of an intracranial pressure monitor, and use of medications to control cerebral perfusion pressure), length of stay, and complications. Primary outcome assessed was functional outcome at discharge and 6 months postdischarge as measured by both Glasgow Outcome Scale (GOS) score¹⁹ and a modified FIM score²⁰ (Table 1). Preinjury FIM score was obtained retrospectively from the patient or family at the time of enrollment into the study. The GOS measures global functioning as a combination of neurologic functioning and dependence on others with five outcome categories ranked 1 through 5: death (1), persistent vegetative state (2), severe disability (conscious but dependent on others for daily activities) (3), moderate disability (disabled but independent in daily activities) (4), and good recovery (5).

The modified FIM measures disability or need for assistance in three areas of activity. These three areas were chosen from the 18 items in the full FIM, one each from the domains of self-care, motor function, and cognitive function. The three items, expression, feeding, and locomotion, each have four possible levels of function ranging from total dependence (1) to independence (4). Total modified FIM scores therefore can range from 3 (total dependence) to 12 (total independence). The use of subgroups from the FIM has been demonstrated to track changes in functional status as well as the total FIM in both brain tumor and TBI patients.^{21,22} Specific components that have shown the most reliability have been those of locomotion and activities of daily living. In addition, the modified FIM has been considered essential as a trauma registry item²³ and has been shown to be useful in tracking functional outcome after TBI in patients entered in to the New York State trauma registry.⁷ Lastly, excellent reliability

Table 1 Modified FIM Scoring 7,23

Expression	
(1) Total Dependence	Dependent with Total Help: Does not express basic needs even with an augmentative device
(2) Partial Dependence	Dependent with Partial Help: Requires cues of another person to express thoughts
(3) Device Dependence	Independent with Device: Communicates basic needs and wants without difficulty. May require an augmentative communication device
(4) None: Independence	Independent: Expresses complex ideas intelligibly and fluently
Feeding	
(1) Total Dependence	Dependent with Total Help: Requires maximal assistance for feeding, relies on other means of alimentation such as gastrostomy tube
(2) Partial Dependence	Dependent with Partial Help: Takes full meals by mouth (chews and swallows) but requires supervision or assistance
(3) Device Dependence	Independent with Device: Requires prior preparation or assistive device
(4) None: Independence	Independent: Eats in customary manner
Locomotion	
(1) Total Dependence	Dependent with Total Help: Requires maximal assistance to go 100 feet
(2) Partial Dependence	Dependent with Partial Help: If walking or using wheelchair, requires supervision
(3) Device Dependence	Independent with Device: Walks a minimum of 150 feet but uses a device such as brace (orthosis), prosthesis, cane, crutches, or walker; operates a wheelchair independently
(4) None: Independence	Independent: Walks a minimum of 150 feet

Total modified FIM score is the sum of the three components and ranges from 3 to 12.

between scores obtained by in-person and phone interviews have been shown using the full FIM score in patients after stroke.²⁴ Thus, the modified FIM was chosen in this study for its ease of use and the ability to conduct follow-up evaluations by phone.

Data were collected by specially trained trauma registry or research personnel (nurses, physician assistants). The data were entered into a relational database designed for the study (Paradox 7.0, Corel Corporation, Ottawa, Ontario) and analyzed using SPSS (SPSS, Inc., Chicago, IL). Data are presented as percentages for categorical data and means with 95% confidence intervals for ordinal and continuous data. Statistical analysis was performed using contingency analysis (χ^2) for categorical data and Student's *t* and Mann-Whitney

Table 2 Mechanism of Injury

	Total	18–64 yr	≥65 yr
Assault	65	64	1
Blunt	8	6	2
MCC	6	6	0
Fall	80	50	30
MVC	57	48	9
Other	15	13	2

MCC, motorcycle crash; MVC, motor vehicle crash.

tests (as indicated) for continuous data. Calculated significance values are reported.

RESULTS

Over the study period (January 2000–July 2002), 244 patients were enrolled. Consent was obtained in 9 patients that died before discharge from the hospital, leaving 235 patients with isolated TBI evaluable for functional outcome. Using data from trauma registries at the four centers, the enrollment rate for the study was approximately 30%. Reasons for nonenrollment included patient under custody, translator not available, and patient discharged before being approached for consent. Contingency analysis and univariate statistics were calculated to look for important differences in demographic and clinical data points. No striking differences among the centers were found except for the expected variation in the racial and ethnic populations and mechanism of injury served by the four trauma centers.

Of the 235 enrolled, 44 (19%) were considered as elderly (age > 65 years) and 187 (81%) were considered as younger adults (age 18–64 years). Forty-nine percent of the study population were white, 22% Hispanic, 18% black, and 10% Asian. A higher percentage of the elderly were women (46%) than the young adults (19%) ($p = 0.003$). Mechanism of injury was most likely to be fall in the elderly group compared with assault in the young adults (Table 2). Mean GCS score of the entire population was 12.8 (95% confidence interval [CI], 12.4–13.3). The mean GCS score of the elderly group was 14.1 (95% CI, 13.6–14.6), which was significantly higher than that of younger adults, at 12.5 (95% CI, 12.0–13.1, $p = 0.03$). As seen in Figure 1, only 4 of 44 (9%) elderly enrolled had a GCS score less than 13 as compared with 40 of 182 (21%) young adults.

Despite this finding, a similar percentage of elderly and young patients were admitted to the intensive care unit (41% young and 47% elderly). There were also no obvious differences in the overall management of the TBI between the groups, with 73% of young and 61% of elderly ($p = 0.94$) receiving any intervention (Table 3). Of the 44 elderly patients, 73% ($n = 32$) had preinjury medical conditions, predominantly hypertension ($n = 22$) and cardiac disease ($n = 16$) compared with only 28% of the young. The in-hospital complication rate was similar between groups (young, 15%; elderly, 18%), as was the length of stay (young, 7.9 days;

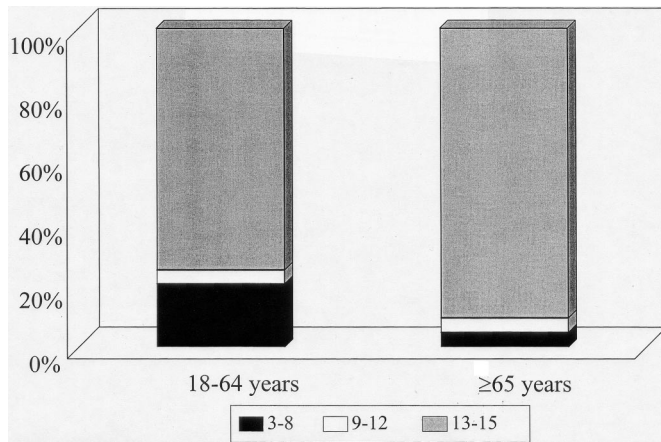


Fig. 1. Glasgow Coma Scale score distribution by age groups: young (18–64 years) and elderly (65 years or older). Data clearly demonstrate that few moderate to severe elderly head-injured patients survive to discharge, and the majority of survivors are those patients with GCS scores of 13 to 15.

elderly, 7.3 days). There were also no differences in discharge disposition between the young and elderly, with 76% of young patients being discharged to home and 23% to inpatient rehabilitation compared with 61% home and 30% to rehabilitation in the elderly group.

Analyzing the entire patient population, there was no difference between elderly and young in functional outcome as measured by GOS at discharge or at 6-month follow-up (young mean GOS score of 4.5 vs. elderly mean GOS score of 4.3). There was a statistically significant difference in the mean FIM score at discharge, which was 10.6 (95% CI, 10.3–11.0) for the young and 9.8 (95% CI, 9.1–10.7) for the elderly ($p < 0.001$). Even so, 61% of the elderly scored 11 to 12 (total independence). A consistently small difference was also noted in the FIM score at 6 months after injury: young, 11.6 (95% CI, 11.5–11.8); elderly, 10.9 (95% CI, 10.6–11.4) ($p < 0.001$).

As there were only a few elderly patients with moderate to severe TBI (GCS score ≤ 12) who survived to discharge and were eligible to be enrolled, meaningful analysis of functional outcome of this subgroup was not possible. Fur-

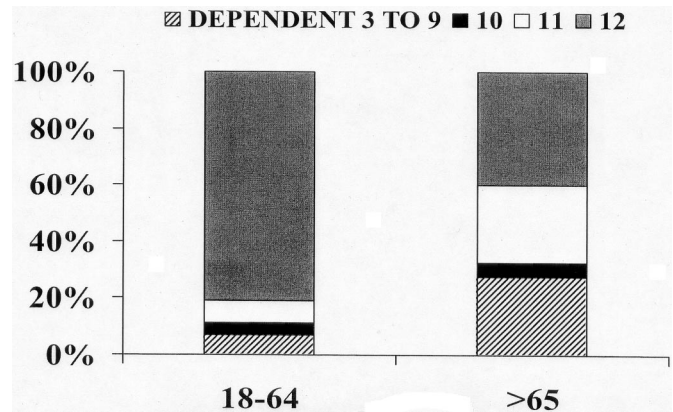


Fig. 2. Discharge FIM scores of patients with GCS scores of 13 to 15. Although most patients demonstrate a high degree of independence (FIM scores of 11 or 12), there is a statistical difference between young (84%) and older (68%) groups.

thermore, the disparity of TBI severity between the young and old patients could have potentially skewed the outcome data in the entire study group toward a more favorable outcome in the elderly. Therefore, we reanalyzed the data with respect to functional outcome only on the mild TBI group (GCS score of 13–15).

In the mild TBI group, there were 182 patients, with 142 young and 40 elderly. Although a greater percentage of elderly were admitted to the intensive care unit (47% vs. 29%, $p = 0.03$), the rate of neurosurgical intervention was similar between the young and elderly groups (59% vs. 48%, $p = 0.92$). Thus, there did not appear to be significant treatment bias between the groups. A greater percentage of elderly were discharged to rehabilitation (28% vs. 16%), although this was not statistically significant ($p = 0.08$). The same small difference in discharge FIM score in the entire study group was observed in the mild TBI patients as well. The mean discharge FIM score was 10.4 (95% CI, 9.8–11.0) for the elderly versus 11.4 (95% CI, 11.1–11.7) for the young ($p = 0.001$), with 68% elderly and 89% young discharged with a total independent score of 11 to 12 (Fig. 2). Elderly patients with “mild” TBI were also more likely to be discharged with GOS score of severe disability than the young (23% vs. 9%, $p = 0.03$); however, the majority (63%) still had a GOS score of 5, consistent with good recovery (Fig. 3).

One hundred twenty-eight patients (70%) in the mild TBI group were evaluable at 6 months for follow-up (2 patients had died and 52 (28%) were lost to follow-up). The mean FIM score was 11.7 (95% CI, 11.6–11.9) for the young and 11.0 (95% CI, 10.6–11.4) for the elderly ($p < 0.001$) (Fig. 4). GOS scores were similar to discharge scores (Fig. 5). When change in function was measured using preinjury FIM as the baseline, greater than 60% of both young and elderly were functioning at preinjury levels, but a greater percentage of the elderly reported a decreased functional outcome (Table 4). When change in functional status was measured using

Table 3 Neurosurgical Interventions by Age Groups

	18–64 yr	≥65 yr
Endotracheal intubation	28	6
Intracranial pressure monitor	16	2
Mannitol	12	0
Pressors for CPP control	5	0
Anticonvulsant therapy	120	23
Craniotomy	10	2
Patients with any intervention*	137	27

CPP, cerebral perfusion pressure.

* Patients can have more than one intervention; thus, there are more interventions than patients.

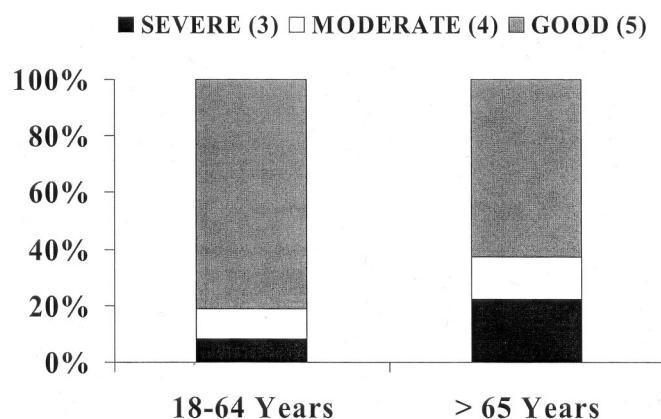


Fig. 3. Discharge GOS scores of patients with a GCS score of 13 to 15.

discharge FIM as control, the elderly were more likely to improve their score (Table 5) than the young, but this was not statistically significant ($p = 0.39$).

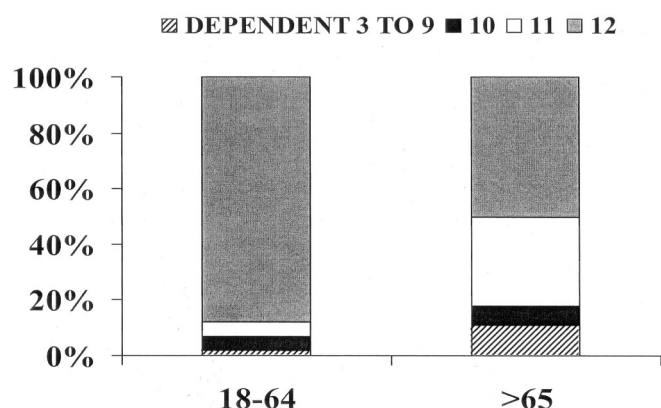


Fig. 4. Six-month postinjury FIM scores of patients with GCS scores of 13 to 15. Note that there is improvement in both groups, although older patients continue to lag behind the younger cohort.

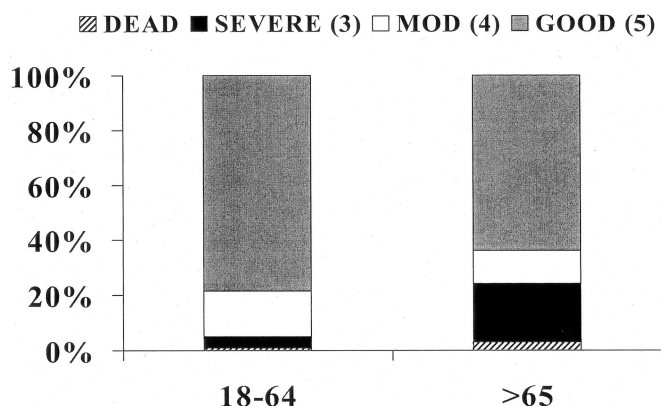


Fig. 5. Six-month GOS scores of patients with GCS scores of 13 to 15.

Table 4 Change in FIM from Preinjury to 6-Month Follow-Up (GCS Score of 13–15)

	All (%)	18–64 yr (%)	≥65 yr (%)
Worse	22 (17)	11 (11)	11 (34)*
Unchanged	104 (81)	84 (88)	20 (63)
Improved	2 (2)	1 (1)	1 (3)

* $p = 0.02$, elderly vs. young.

DISCUSSION

Although survival after injury is the “ultimate” outcome, the quality of life of patients who do survive their initial trauma is probably of greater importance. As traumatic brain injury remains one of the most significant health problems treated in trauma centers, any data on the functional outcome of patients sustaining isolated TBI would be helpful to both providers and patients. The results presented here demonstrate several important findings. First is that the majority of older patients who sustain mild TBI do quite well and are able to live independently. Although there were statistically significant differences in the FIM score at both discharge and 6 months, the clinical importance of these small differences in the mean FIM score to the patient’s quality of life is less clear. The second is that there is measurable improvement in functional status during the first 6 months after injury. Lastly, the data further document that survival after isolated moderate and severe TBI in older patients is significantly lower than in younger patients. On the basis of this study, we conclude that functional outcome after isolated mild TBI as measured by the GOS and the modified FIM is generally good to excellent for both elderly and younger patients. Older patients required more intensive care unit resources and inpatient rehabilitation and lagged behind their younger counterparts, but they continued to recover and improve after discharge. As increasing numbers of elderly are hospitalized after trauma and TBI, evidence of equivalent functional outcome justifies intensive resource use for their care both in the acute and in the rehabilitation hospital settings.

The expected functional outcome of older patients after TBI is difficult to ascertain from the literature. Significant limitations include retrospective study design, small sample sizes, inclusion bias that evaluated only patients referred to inpatient rehabilitation, and the presence of multiple other injuries that may have a greater impact on final outcome than the brain injury itself.^{11,15,20} This study attempted to address these shortcomings. The goal was to prospectively enroll all patients with isolated TBI, therefore eliminating both the

Table 5 Change in FIM from Discharge to 6-Month Follow-Up

	All (%)	18–64 yr (%)	≥65 yr (%)
Worse	8 (6)	5 (5)	3 (9)
Unchanged	95 (74)	75 (78)	20 (63)
Better	25 (20)	16 (17)	9 (28)

retrospective nature of the previous studies and any potential inclusion bias. The data from the trauma registries demonstrate that although we were only partly successful in eliminating enrollment bias, overall, approximately one third of all potential patients were enrolled. Improving enrollment would have also provided more than the 40 elderly patients on which to base our conclusions. The predominant reasons why eligible patients were not enrolled were either a language barrier or a short stay (<2 days). There were no differences in the demographics or the percentage of patients that were eligible but not enrolled by age. Thus, although enrollment could have been more complete, there did not appear to be any systematic bias that resulted in our study group not being representative of the population as a whole.

Lower FIM scores at discharge from the hospital in elderly patients after TBI have been reported by others, but this difference disappeared at 6-months or 1-year follow-up, after rehabilitation was completed. In addition, the elderly had an increased need for inpatient rehabilitation and longer rehabilitation lengths of stay to achieve this goal.^{12,15} Similarly, in the present study, a higher percentage of elderly improved their functional outcome from discharge to 6-month follow-up than did the young. Whether this is related to admission to rehabilitation, which was higher among the elderly, or merely reflects a slower rate of progress in functional measures is unclear. Although a preinjury FIM score was obtained on all patients, additional comorbidities that are more likely to be present in the elderly were not captured and may account for this finding. Length of stay in rehabilitation and ongoing exposure to various therapeutic modalities has been shown to correlate with improvements in functional outcome after stroke and brain tumor.²¹ It also remains possible that the lower initial scores of the older patients merely allowed for greater improvement. Further longitudinal study of this population may reveal further improvement in FIM and GOS scores among the elderly, supporting earlier literature.

The relatively good functional outcome for the elderly seen in this study may be explained by several unique features of the study population. Most importantly, this study investigated patients with isolated TBI. The majority of studies that found elderly functional outcome to be poor included patients with inadequately defined multiple injuries as well as TBI.^{7,11} The overall impact of multisystem injuries, especially lower extremity injuries, on functional outcome and quality of life has clearly been shown to be significant,^{25,26} with long-term improvements in functional outcome occurring well after 12 months postinjury. Thus, our data suggest that, when multisystem trauma is controlled for, the old and young with pure TBI have fairly similar outcomes at discharge and at 6 months. It is possible that in the setting of TBI plus multiple traumatic injuries, age may play a more critical role in functional outcome because of physiologic interactions of the brain and other injuries as well as possible differences in treatment.

Second, this study is focused predominantly on patients sustaining mild TBI. It is possible that age becomes a more important predictor of a poor functional outcome when TBI is more severe.¹¹ Others have suggested that cognitive deficits after TBI may be more important to long-term function for the elderly, and these may be more apparent in the severely injured patient. Lastly, we divided the elderly group at age 65 based on our prior work in an increased mortality after TBI.⁵ It is more likely that the effect of age on recovery changes continuously throughout life rather than at an arbitrary time point. This stepwise effect of increasing age on poorer outcome was reported in a recent meta-analysis of patients sustaining TBI.¹⁰ Increased numbers of patients, especially those over 70 and 80 years, would be needed to truly assess the continuous effect of age on recovery from TBI. The follow-up rate for this study, both for the elderly and for the young, suggests that it is unlikely that selective loss to follow-up of lower functioning adults accounts for the overall good functional outcome.

Although there were statistical differences in FIM scoring between younger and older groups, and the majority of elderly had independent function assessed by global functional measures, it is possible that more clinically significant differences in outcome between the elderly and the young were present, but the instruments used did not detect them. The GOS, although useful, provides only a global assessment of function and dependence, but may not differentiate specific differences in cognitive function, motor function, or activities of daily living. The modified FIM selects only 3 items from the 18-item score, and also distinguishes only four (as opposed to seven) levels of function. Subtle or complex deficiencies, particularly in cognitive function in the elderly, may not have been identified in the data set. However, Huang and colleagues showed that changes in the FIM subscores, particularly the motor and activities of daily living scores correlated well with changes in the total FIM score in 78 patients admitted to inpatient rehabilitation.²¹ O'Dell et al. divided the FIM into six subcategories and found similar improvements in motor and activities of daily living subscores as compared with the entire FIM. A lag in improvement in the cognition components in the FIM compared with motor and activities of daily living portions of the FIM have been shown in TBI, stroke, and brain tumor patients²² and may not have been observed at discharge or at 6 months.

A deficiency in using any one scale to measure outcome is that it is limited in its scope of measurement. Although we attempted to address what we felt were clinically important issues, these scales did not assess for posttraumatic stress disorder, depression, or other aspects of quality of life, all of which may reveal important differences for the elderly versus the young. In addition, there is always a concern about interrater reliability when performing multicenter trials. All individuals assigning the patients GOS and FIM scores underwent instruction by the principal investigator at each site and thus we do not believe there were any great differences in the

reliability of the scores within or between sites. Previous studies, although retrospective, used the 18-item FIM, including more cognitive measures and/or the Disability Rating Scale,^{12–15} and have noted differences in elderly function. It is possible that cognitive function may be more important for overall outcome in the elderly than in the young, which our instruments did not measure. Although there were no gross differences in the severity of head injury as measured by the GCS in those patients sustaining mild TBI, we did not analyze outcome with respect to lesion location. In studies of stroke and brain tumor, location of lesion has been demonstrated to influence outcome.²¹ Similar data on traumatic injury are less clear. Furthermore, there were differences in the mechanism of injury between the groups. It is possible that both the mechanism of injury and the social and economic circumstances of the victims influence the severity of the brain injury as well as the functional recovery.²⁷

In conclusion, the majority of patients lived independently after their isolated mild TBI, and age does not appear to be a limiting factor with respect to eventual functional outcome. As few elderly patients with moderate to severe TBI who survived to discharge were enrolled, the eventual outcome of this cohort unfortunately remains unknown. Elderly patients with mild TBI had consistently lower FIM scores at discharge and at 6 months than their younger counterparts; however, the clinical implication of these differences is difficult to determine from our data. These differences also appeared to narrow from discharge to 6 months. Completion of this prospective study with follow-up at 1 year postinjury may reveal further insights into the role of age and other factors that influence the outcome of mild TBI. Future prospective studies, possibly with more detailed and varied instruments and attention to cognitive elements, posttraumatic stress disorder, and quality of life, are clearly needed to further assess the long-term functional outcome of TBI.

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